

Unit 3

HARDWARE

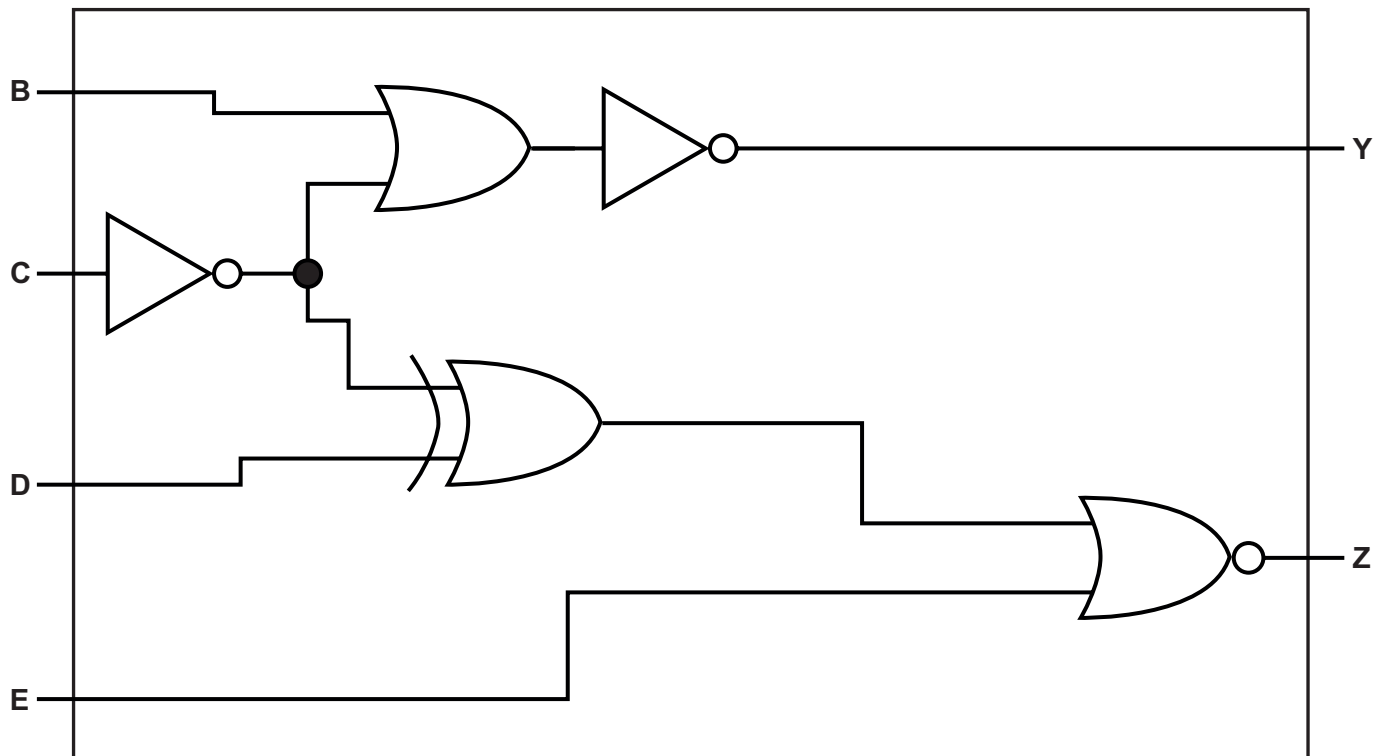
61 items · 307 marks

Syllabus sections in this unit

- 3.1 Computers and their components
- 3.2 Logic Gates and Logic Circuits

Unit 3: Hardware**1. M/J 26/P11/Q6/a,b**

6 (a) Write the logic expressions for the following logic circuit.



Y =

Z =

[2]

(b) Create and complete the truth table for a two-input NAND gate.

[2]

2. M/J 26/P12/Q6/a,b,c

- 6** A health monitor attaches to a person's arm and measures key health information, for example heart rate. The monitor transmits data to the user's smartphone. The results are analysed by the phone, which displays information to the user about their health.

(a) Explain the reasons why the health monitor is an example of an embedded system.

[3]

(b) Tick (✓) **one** box to identify if the health monitor is an example of a monitoring system or a control system.

system	tick (✓)
monitoring	
control	

Justify your choice.

[2]

(c) The health monitor contains Random Access Memory (RAM) and Read Only Memory (ROM).

Describe **two** differences between RAM and ROM.

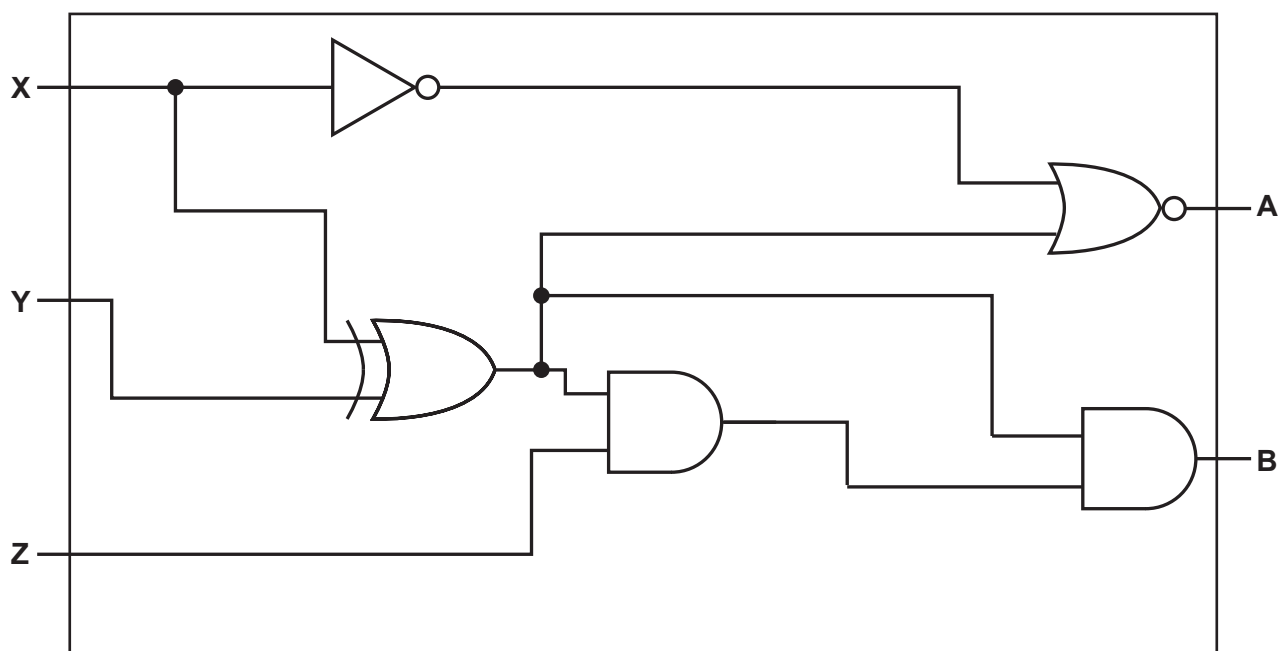
1

2

[2]

3. M/J 26/P12/Q8/a,b

- 8** (a) Complete the truth table for the given logic circuit.



X	Y	Z	working space	A	B
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

[2]

(b) Write the logic statement for the given truth table.

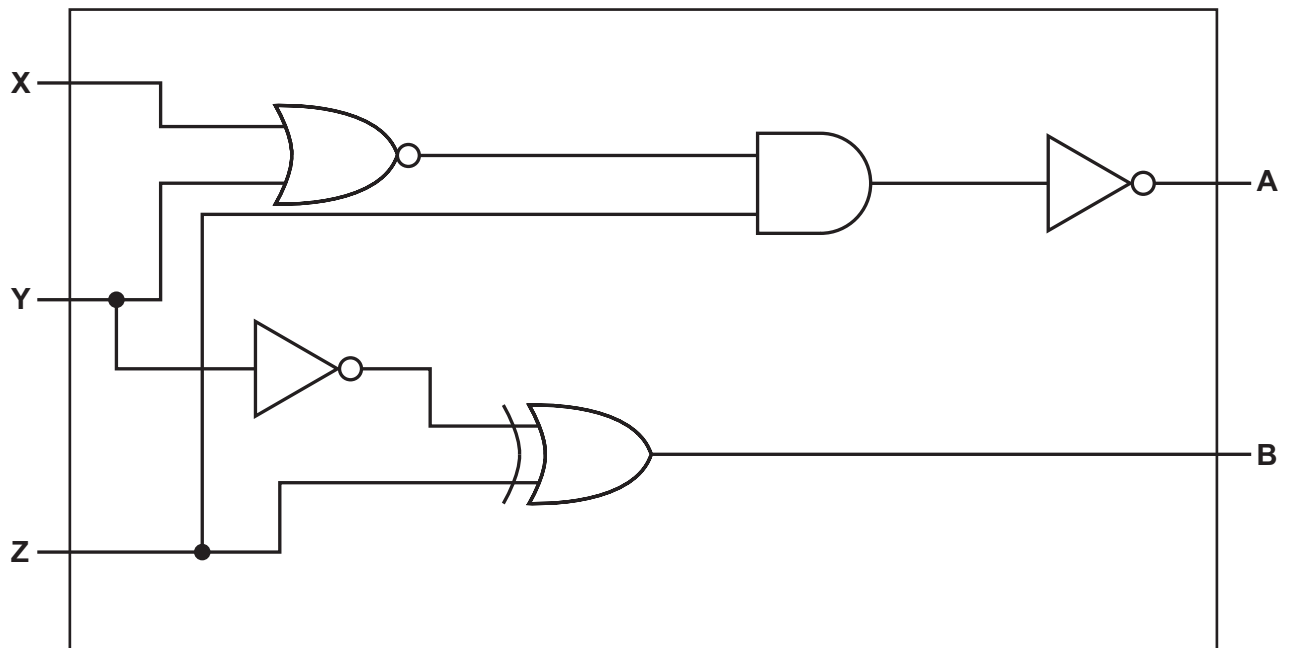
input			output
A	B	C	P
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

P =

[1]

4. M/J 26/P13/Q2/a,b

2 (a) Complete the truth table for the given logic circuit.



X	Y	Z	working space	A	B
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

[2]

(b) State **two** logic gates that are not in the circuit in **2(a)**.

[1]

5. M/J 26/P13/Q8/a,b,c

8 A computer is used to record sound.

(a) The computer has Programmable Read Only Memory (PROM).

Complete the table by naming and describing two other types of ROM.

type of ROM	description

[4]

(b) The sound is recorded using a microphone.

Describe the principal operation of a microphone.

[3]

(c) The computer outputs the sound through external speakers. The computer transfers the data to a buffer before the speakers output the sound.

Describe the purpose of the buffer.

[3]

6. O/N 25/P11/Q3/a,b

3 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT} ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$



[2]

(b) Write the logic expression for the following truth table.

R	S	T	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Q =

[2]

7. O/N 25/P11/Q6/c

(c) State **three** differences between Dynamic RAM (DRAM) and Static RAM (SRAM).

- 1
- 2
- 3

[3]

8. O/N 25/P11/Q9

- 9 Describe the differences between a monitoring system and a control system.

[3]

9. O/N 25/P11/Q10/a,b

- 10 An architect needs a model of a building.

(a) Explain how the model will be printed using a 3D printer.

[3]

(b) State why a 3D printer needs a buffer.

[1]

10. O/N 25/P12/Q8/b(i)

- 8 A restaurant accesses software and stores data using cloud computing.

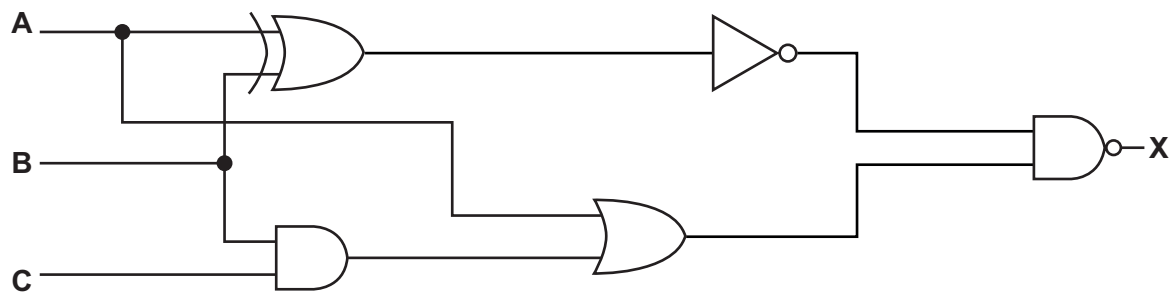
(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(i) Explain how a touchscreen converts the point of touch to a selection from the menu.

[4]

11. O/N 25/P12/Q9/a,b

9 (a) Consider the following logic circuit:



Write the logic expression for the logic circuit. Do **not** simplify the expression.

X =

[2]

(b) Consider the following logic expression:

$$X = \text{NOT } ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$

A truth table for the logic expression is given:

Row number	A	B	C	X
1	0	0	0	1
2	0	0	1	0
3	0	1	0	1
4	0	1	1	0
5	1	0	0	1
6	1	0	1	0
7	1	1	0	1
8	1	1	1	1

There are **three** errors in the truth table.

Identify the **three** errors in the truth table by writing the row numbers with an incorrect output.

Error 1 Row number

Error 2 Row number

Error 3 Row number

[2]

12. O/N 25/P12/Q11

- 11** An automated system opens doors when a person is detected within 2 metres. The system closes the doors when there is no longer a person within 2 metres.

Identify whether the automated system is an example of a monitoring system or a control system.

Justify your choice.

Monitoring or Control

Justification

[3]

13. O/N 25/P13/Q3/a,b

3 (a) Complete the truth table for the following logic expression:

$$X = (A \text{ NOR } B) \text{ NAND } (C \text{ XOR } B)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

$$X = \text{NOT } (((A \text{ AND } B) \text{ OR } C) \text{ AND } (C \text{ NOR } D))$$



[2]

14. O/N 25/P13/Q4/c

- 4 A network engineer is setting up a Local Area Network (LAN) in an office. The network uses both wired and wireless connectivity.

(c) Complete the table by writing the name of the type of memory technology for each description.

Description	Memory technology
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	
A read only memory where the contents can be written only once after manufacture.	

[3]

15. O/N 25/P13/Q7/d

- 7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(d) Describe **two** reasons why the server that stores the videos uses magnetic hard disks instead of solid state (flash) memory.

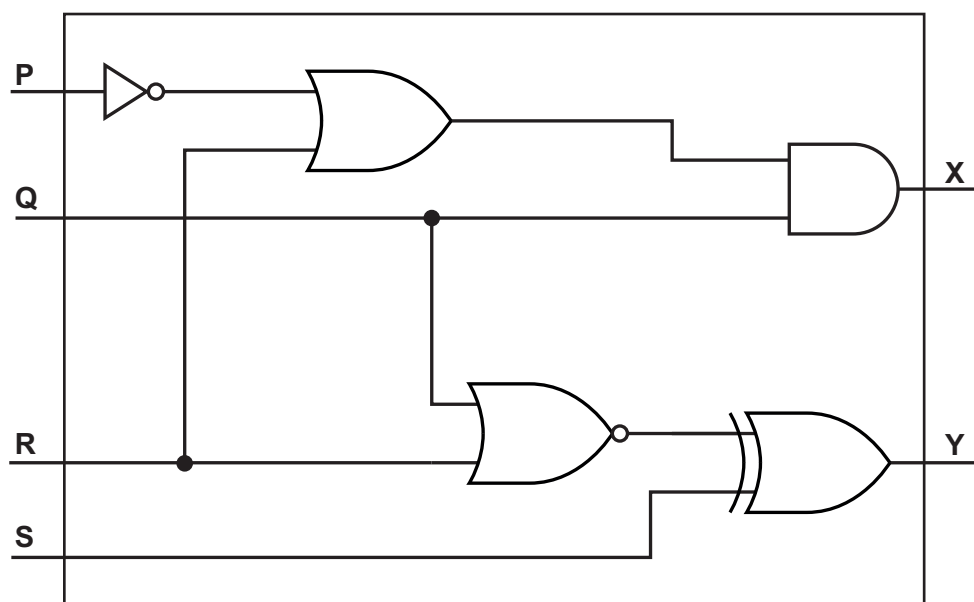
1

2

[4]

16. M/J 25/P11/Q1/a,b

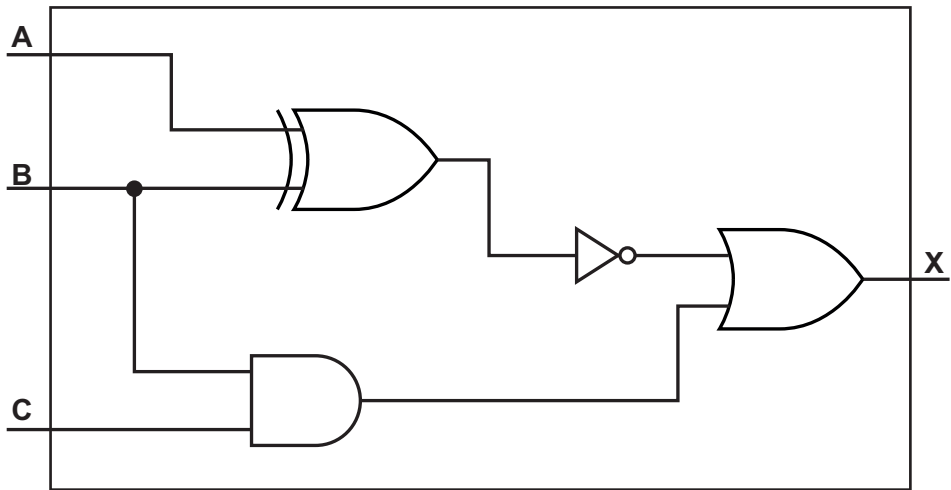
- 1 (a) Write the logic expressions for the following logic circuit.



X =
Y =

[2]

(b) Complete the truth table for the following logic circuit.



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

17. M/J 25/P11/Q4/b

- 4 A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer's account.

- (b) The new system uses sensors to identify the items taken from a shelf.

Identify **one** type of sensor that can be used in this new system.

State how the sensor can be used to identify the items taken from a shelf.

Sensor

Use

[2]

18. M/J 25/P11/Q6/a(i)

- 6 A programmer is buying a new computer.

- (a) The programmer is considering the following two computers:

Computer 1	Quad-core 2.2 GHz processor 16-bit architecture 1 GB Random Access Memory (RAM) 500 GB magnetic hard disk
Computer 2	Dual-core 3.8 GHz processor 32-bit architecture 2 GB RAM 500 GB solid state drive (SSD)

- (i) Computer 1 has a magnetic hard disk.

Complete the description of the principal operation of a magnetic hard disk by writing the missing words.

The magnetic hard disk has one or more that can be magnetised. These are mounted on a and rotate at high speed.

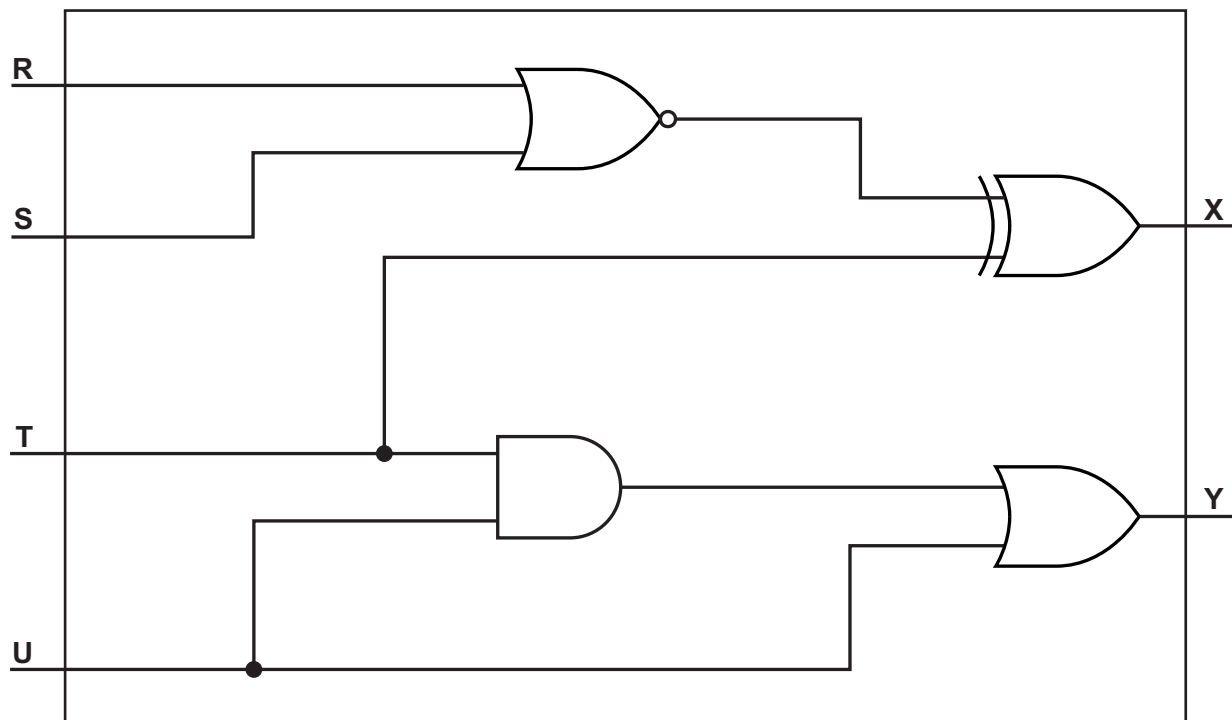
A is moved across the surface on an arm.

When data is read, the changes in the produce a change in the electric current.

[4]

19. M/J 25/P12/Q4/a,b

4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

[2]

(b) Complete the truth table for the logic expression:

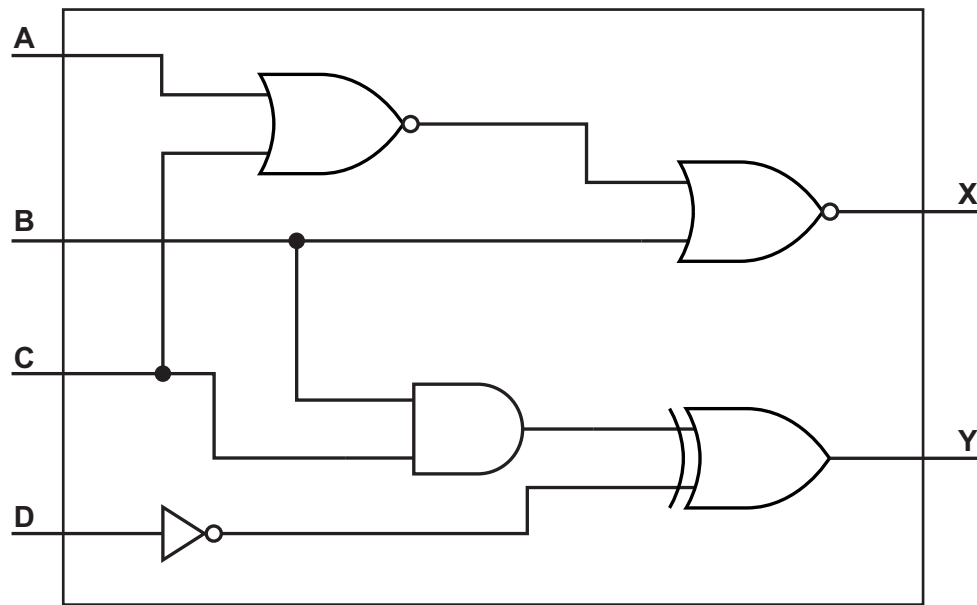
$$X = \text{NOT } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

20. M/J 25/P13/Q4/a,b

4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

[2]

(b) Complete the truth table for the logic expression:

$$X = (A \text{ OR } B) \text{ XOR } (B \text{ OR } C) \text{ XOR } (\text{NOT } A \text{ NAND } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

21. O/N 24/P11/Q3/c,d

3 A computer system has a dual-core Central Processing Unit (CPU).

(c) Identify **two** disadvantages of using Dynamic RAM (DRAM) instead of Static RAM (SRAM) in a computer system.

1

2

[2]

(d) The computer system is used to store data received from a temperature sensor every five seconds. The data is stored on an optical disc using an optical disc reader/writer.

(i) Describe the principal operation of an optical disc reader/writer.

[4]

(ii) The computer uses a buffer when writing data to the optical disc.

Explain the use of a buffer when writing data to the optical disc.

[3]

22. O/N 24/P11/Q5/a,b,c

5 A security system has both a floodlight (very bright light) and an audio alarm.

The data from multiple sensors is analysed and used to:

- turn on the floodlight
- sound the audio alarm.

Sensors can be used to detect:

- if doors are open
- the external daylight level
- if people are detected within a set distance.

(a) Complete the table to identify the most appropriate type of sensor for each scenario.

Scenario	Type of sensor
A door is open.	
The external daylight level is below a set amount.	
A person is detected within 2 metres.	

[1]

(b) The floodlight (**X**) and audio alarm (**Y**) operate according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
A	external daylight level	1	Low
		0	High
B	front door	1	Open
		0	Closed
C	person is within 2 m	1	Detected
		0	Not detected
D	back door	1	Open
		0	Closed
E	security system	1	Switched on
		0	Switched off

The floodlight turns on (**X = 1**) if:

- the security system is switched on
and
- the external daylight level is low
and
- a person is detected within 2 m.

The audio alarm turns on (**Y = 1**) if:

- the security system is switched on
and
- one **or** more doors are open, **or** a person is detected within 2 m.

Write logic expressions for the security system.

X =

Y =

[2]

(c) Explain whether the security system is an example of a monitoring system or a control system.

[3]

23. O/N 24/P12/Q1/a,b

1 (a) Complete the truth table for the logic expression:

$$X = (A \text{ XOR } B) \text{ NAND } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw the logic circuit for the logic expression:

$$W = P \text{ NAND } ((Q \text{ OR NOT } R) \text{ XOR } (P \text{ XOR } Q))$$



[2]

24. O/N 24/P12/Q2/a,b,c

2 Embedded systems are used in many electronic devices.

(a) Describe the drawbacks of embedded systems.

[3]

(b) An embedded system has Dynamic RAM (DRAM).

Identify **one** benefit of using DRAM instead of Static RAM (SRAM) in an embedded system.

[1]

(c) Give **two** differences between Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM).

Difference 1

Difference 2

[2]

25. O/N 24/P12/Q9/a,b

9 A road bridge has a weight limit and a height limit for vehicles. For example, a vehicle must weigh less than 10 000 kg and must have a height of less than 3 m.

The bridge has a warning system. If a vehicle is approaching the bridge and it exceeds one or both limits, a sign displays a warning telling the driver of the vehicle to stop.

(a) The bridge warning system uses sensors to detect if a vehicle exceeds the limits.

Complete the table by identifying **two** different sensors that could be used by the system **and** describe how each sensor is used by the system.

Sensor	Use in bridge warning system
.....	
.....	

[2]

(b) Explain whether the bridge warning system is an example of a monitoring system or of a control system.

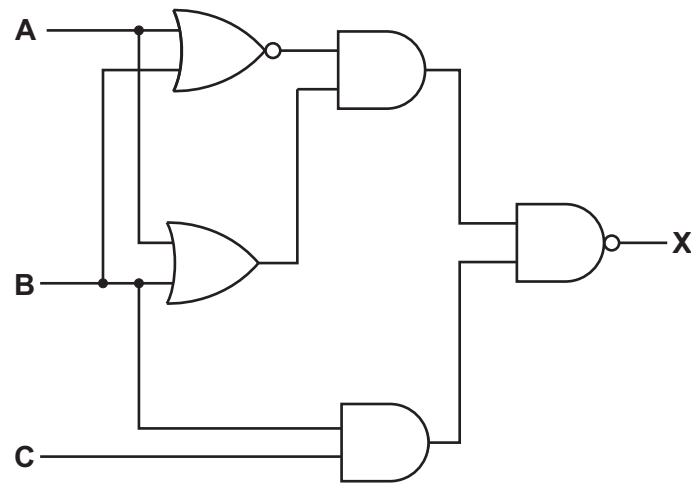
[2]

26. O/N 24/P13/Q1/a,b,c

- 1 (a) Describe the operation of a 2-input XOR gate.

[1]

- (b) Write the logic expression for the following logic circuit.



X =

[2]

- (c) Draw the logic circuit for the logic expression:

$$X = ((\text{NOT } A \text{ AND } (B \text{ AND } C)) \text{ OR } (B \text{ NAND } C)) \text{ AND NOT } A$$



[2]

27. O/N 24/P13/Q2/a,b

2 A memory buffer uses Dynamic RAM (DRAM).

(a) Identify **two** differences between DRAM and Static RAM (SRAM).

Difference 1

Difference 2

[2]

(b) Explain how a memory buffer is used when a computer is transferring data to its magnetic hard disk drive.

[4]

28. O/N 24/P13/Q3

3 A car has an automatic braking system.

A sensor is used to measure the distance to objects that are in front of the car.

The car automatically brakes if an object is too close to the front of the car or the distance between the car and the object is decreasing too quickly. The closer the object is to the front of the car, the harder the car brakes so that the car slows down more quickly.

Explain the reasons why the automatic braking system of the car is a control system.

[3]

29. M/J 24/P11/Q1/a,b

1 (a) Tick (✓) **one** box to identify the correct logic statement for this truth table.

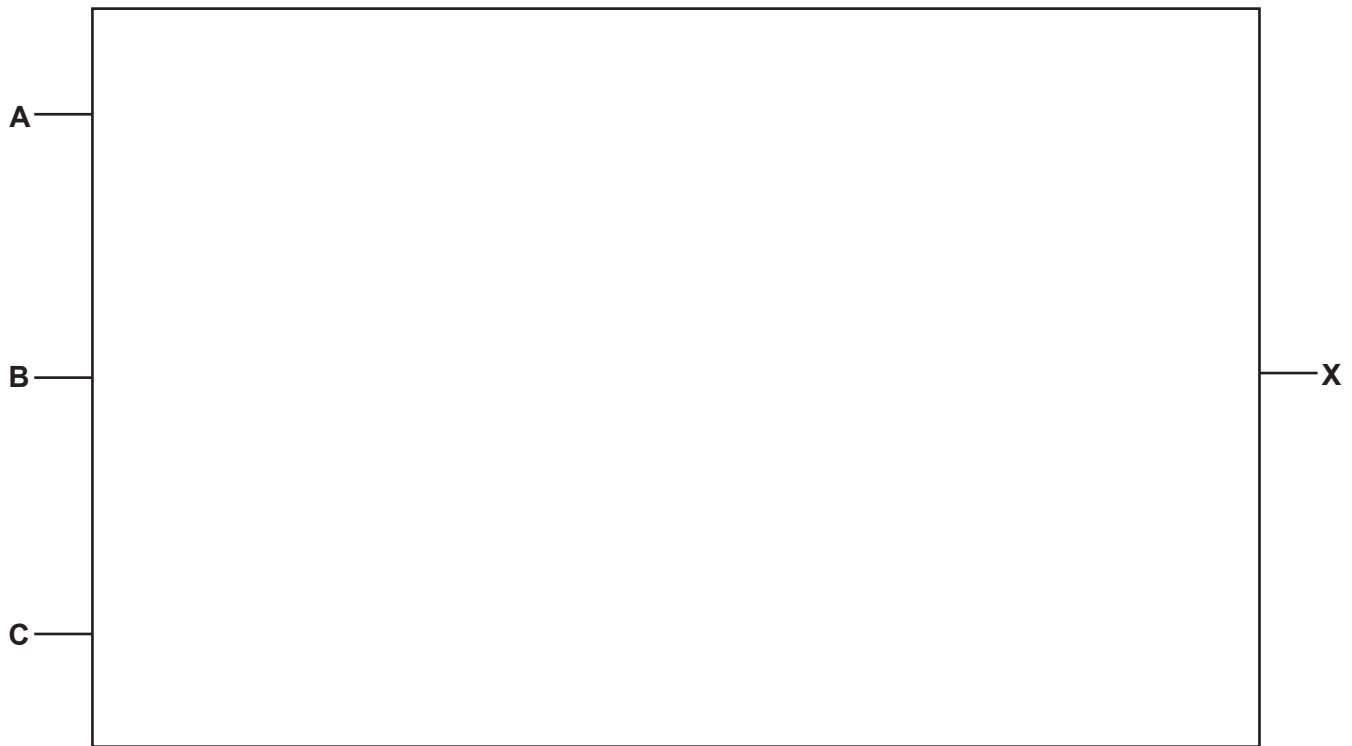
A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

- ☐ NOT (A AND B AND C)
- ☐ (A XOR B) NOR C
- ☐ (A OR B OR C) NOR C
- ☐ NOT A AND NOT B AND NOT C

[1]

(b) Draw a logic circuit for the logic expression:

$$X = \text{NOT} (\text{NOT } A \text{ AND } (\text{NOT } B \text{ XOR } C))$$



[2]

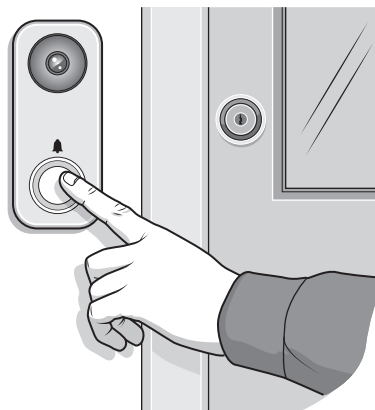
30. M/J 24/P11/Q2/a,b,c

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (a) The video doorbell can be considered an example of an embedded system.

Identify **two** characteristics of the doorbell that suggest it is an embedded system.

2

[2]

- (b) State whether the video doorbell is a monitoring system or a control system. Justify your choice.

Monitoring or control system

Justification

[2]

- (c) The video doorbell has both primary memory and secondary storage.

- (i) Identify **two** items of data that the video doorbell will store in primary memory.

1

2

[2]

- (ii) The video doorbell has a solid state (flash) secondary storage device.

Complete the table by writing the answer or answers to each statement about the principal operation of solid state (flash) memory.

Statement	Answer
the two types of logic gate that can be used to create solid state devices	1 2
the number of transistors contained in each cell	
the type of gate that can retain electrons without power	
the type of gate that allows or stops current from passing through	

[4]

- (iii) The video doorbell uses a buffer.

Describe how the video doorbell will use the buffer.

[2]

31. M/J 24/P12/Q1/a,b

- 1 (a) Describe the operation of each of the following logic gates:

NAND

NOR

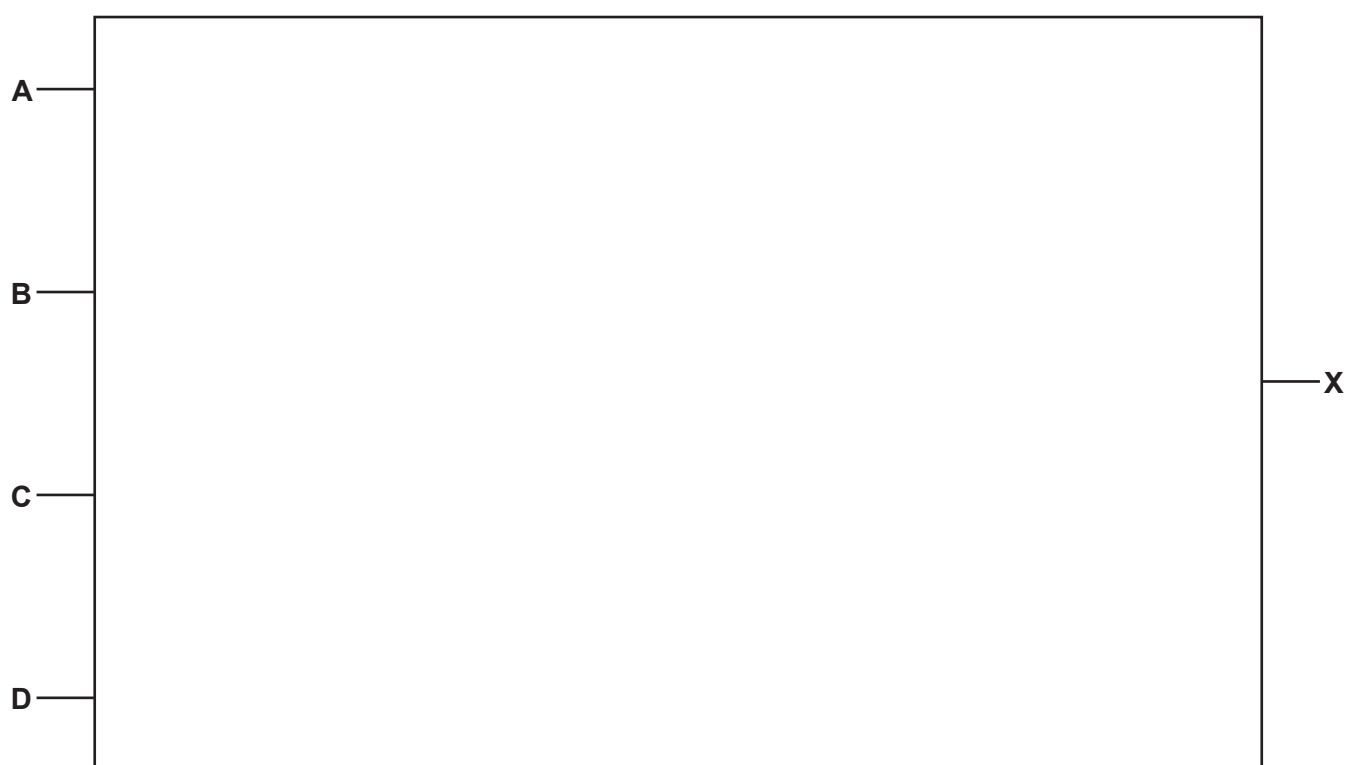
XOR

OR

[4]

- (b) Draw a logic circuit for this logic expression:

$$X = \text{NOT} ((A \text{ AND } B) \text{ OR } (C \text{ AND } D))$$



[2]

32. M/J 24/P12/Q2/a,b,c

- 2** A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(a) Complete the description of the principal operation of a VR headset.

A headset can have one or two that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor.

This sensor is a The data is transmitted to a microprocessor that analyses the data to identify the of movement. Some headsets use that record the user's eye movements for analysis.

[4]

(b) The computer uses a buffer when transmitting data to the VR headset.

Explain how a buffer is used when data is transmitted between the computer and the VR headset.

[3]

(c) The VR headset has Electrically Erasable Programmable Read Only Memory (EEPROM).

Explain the benefits of using EEPROM instead of other types of Read Only Memory (ROM) in the VR headset.

[3]

33. M/J 24/P13/Q6

6 Each of the following truth tables has three inputs (**A**, **B** and **C**) and one output (**X**).

Draw **one** line to match each truth table with its logic expression.

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Logic expression

NOT (**A** XOR **B**) AND **C**

(**A** OR **C**) AND NOT **B**

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A NAND **B** NAND **C**

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

(**A** NAND **B**) OR **C**

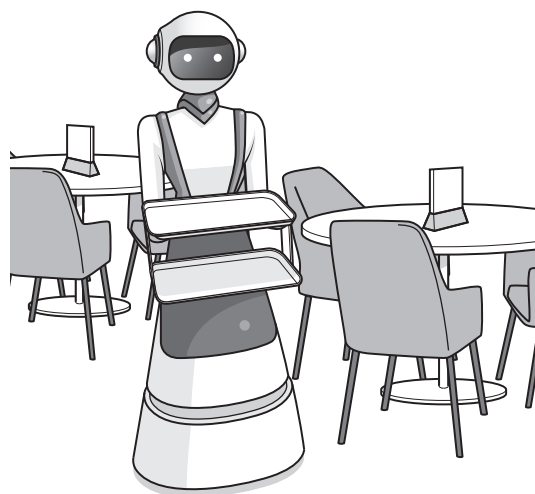
NOT (**A** AND **B** AND **C**)

[3]



34. M/J 24/P13/Q7/a,c,d

7 Robots are used to serve food and drink to customers at a restaurant.



(a) A robot navigates through the restaurant to the table it is serving.

Complete the table by identifying **two** sensors that can be included in the robot **and** the purpose of each sensor in the navigation system.

Sensor	Purpose of sensor in navigation system

[2]

(c) The navigation system can be considered an example of a control system.

Describe how feedback is used in a control system.

[2]

(d) The robot includes a touchscreen for the customer to make their payment.

Describe the principal operation of a touchscreen.

[4]



35. O/N 23/P11/Q4/a,b

4 (a) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ NAND } B) \text{ XOR } (\text{NOT } B \text{ AND } (B \text{ NOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

$$X = (A \text{ AND NOT } (B \text{ OR } C)) \text{ AND } (B \text{ NOR } C)$$



[2]

36. O/N 23/P11/Q7/a,b,c

- 7 (a) Describe the principal operations of a 3D printer.

[3]

- (b) Describe the purpose of a temperature sensor within the 3D printer.

[2]

- (c) A 3D printer contains 1 GB of Dynamic RAM (DRAM) to store print data.

State **two** advantages of the printer having Dynamic RAM instead of Static RAM (SRAM).

1

2

[2]

37. O/N 23/P11/Q9/a,b

- 9 (a) Explain the importance of feedback in a control system.

[2]

- (b) Give **one** example of an embedded system **and** explain why it is an example of an embedded system.

Example

Explanation

[3]

38. O/N 23/P12/Q1/a,b,c

- 1** A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

- (a)** An automated system counts the number of chocolate bars made at the end of production.

The system includes a sensor positioned above the conveyor belt.

Identify **one** appropriate type of sensor that can be used.

[1]

- (b)** A second automated system removes chocolate bars with an incorrect weight from the production line.

Describe the role of an **actuator** in this second system.

[2]

- (c)** The factory has many different machines with embedded systems.

- (i)** Identify **two** features of embedded systems.

1

2

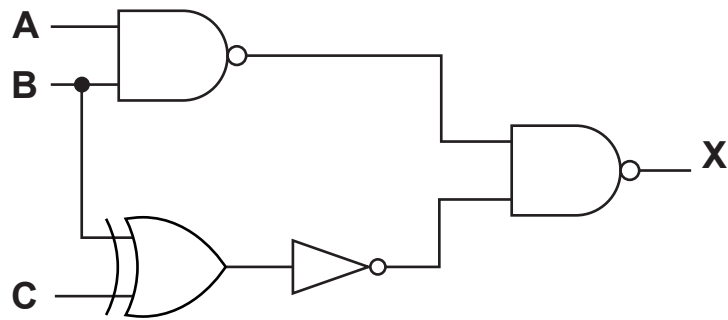
[2]

- (ii)** Identify **one** drawback of embedded systems.

[1]

39. O/N 23/P12/Q4/a,b

- 4 (a) Write the Boolean expression that corresponds to the following logic circuit.



[3]

- (b) Complete the truth table for the logic expression:

$$X = A \text{ XOR } (B \text{ AND } (A \text{ NAND } B) \text{ AND NOT } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

40. O/N 23/P13/Q4/a,b

- 4 (a) Complete the truth table for the logic expression:

$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$



[2]

41. O/N 23/P13/Q7/a,b,c

- 7 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (a) Explain why Static RAM is used in the laptop instead of Dynamic RAM.

[2]

- (b) Identify **two** reasons for using Electrically Erasable Programmable ROM (EEPROM) in a virtual reality headset.

1

2

[2]

- (c) Describe how the laser printer makes use of a buffer.

[4]

42. M/J 23/P11/Q4/a,b

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (a) This computer has both Read Only Memory (ROM) and Random Access Memory (RAM).

- (i) Describe the contents of the ROM in the central computer.

[2]

- (ii) The central computer has Dynamic RAM (DRAM).

Identify **two** advantages of using DRAM instead of Static RAM (SRAM).

1

2

[2]

- (b) The central computer stores the video files on secondary storage.

Describe **two** reasons why magnetic storage is more appropriate than solid state storage for this computer.

1

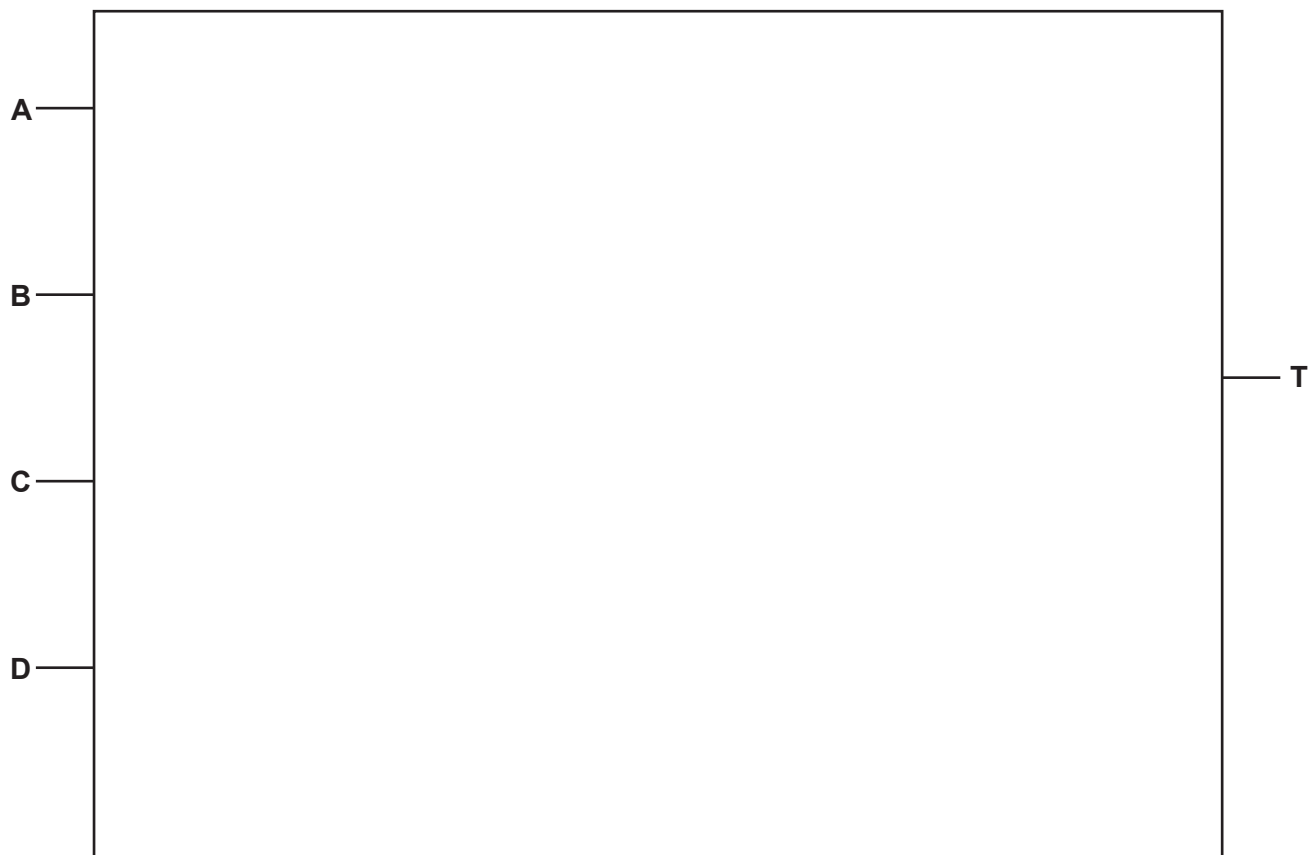
2

[4]

43. M/J 23/P11/Q5/a,b

- 5 (a) Draw the logic circuit for this logic expression:

$$T = (\text{NOT } A \text{ OR } B) \text{ XOR } (C \text{ NAND } D)$$



[2]

- (b) Describe the function of the NAND and NOR logic gates.

NAND

NOR

[2]

44. M/J 23/P12/Q6/a,b

6 (a) Draw the logic circuit for this logic expression:

$Z = (R \text{ XOR } S) \text{ AND } (\text{NOT } T \text{ NOR } P)$



[2]

(b) Complete the truth table for this logic expression:

$Z = (\text{NOT } P \text{ OR } Q) \text{ XOR } (R \text{ NOR } Q)$

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

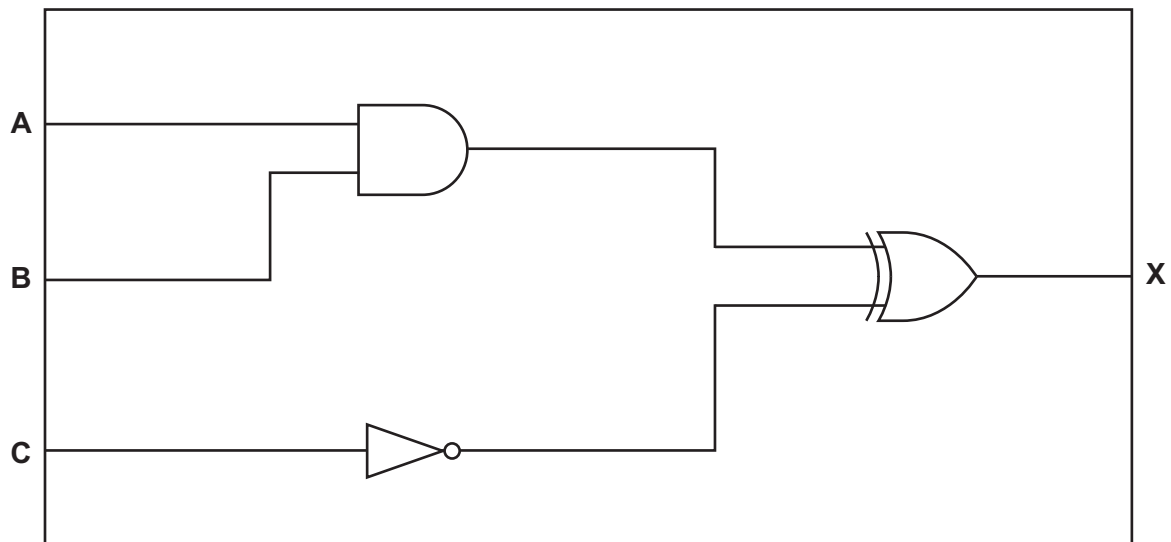
45. M/J 23/P13/Q1/a,b

1 (a) Write the logic expression for this truth table:

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

[1]

(b) Complete the truth table for this logic circuit:



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

46. M/J 23/P13/Q3/a

3 A mobile telephone is used to record a video.

(a) The mobile telephone has a touchscreen. There are different types of touchscreen.

Complete the description of the principal operation of touchscreens.

A touchscreen has two layers. When the user touches the screen, the layers touch and a is completed.

A touchscreen has several layers. When the top layer is touched, there is a in the electric current.

A microprocessor identifies the of the touch.

[5]

47. O/N 22/P11/Q3/a,b

3 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$



[2]

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

48. O/N 22/P11/Q9/a,b

9 Many modern televisions are examples of embedded systems.

(a) Explain why these televisions are embedded systems.

[2]

(b) Embedded systems use Electrically Erasable Programmable ROM (EEPROM).

Describe **one** benefit of using EEPROMs in an embedded system.

[2]

49. O/N 22/P12/Q3/a,b,c

3 (a) A greenhouse has an automatic window.

The window (**X**) operates according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
T	Temperature	1	Too high
		0	Acceptable
W	Wind speed	1	Too high
		0	Acceptable
R	Rain	1	Detected
		0	Not detected
M	Manual override	1	On
		0	Off

The window opens (**X** = 1) if:

- the temperature is too high **and** the wind speed is acceptable
- **and**
- rain is not detected, **or** the manual override is off.

Draw a logic circuit to represent the operation of the window.



[3]

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ OR } B \text{ OR } C) \text{ AND } (B \text{ NOR } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(c) Embedded systems contain Read Only Memory (ROM) and Random Access Memory (RAM).

Explain the reasons why ROM is used in an embedded system.

[2]

50. O/N 22/P13/Q1/c

1 A digital audio message needs to be recorded.

(c) A computer uses a buffer when playing the audio message.

Explain the purpose of a buffer in a computer system using **one other** example.

[3]

51. O/N 22/P13/Q5/a,b

5 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT} ((\text{NOT} (A \text{ AND } B)) \text{ OR } (\text{NOT} (B \text{ AND } C)))$$



[3]

(b) Complete the truth table for the logic expression:

$$Y = (\text{NOT } P \text{ AND } Q) \text{ OR } (Q \text{ AND } \text{NOT } R)$$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

52. O/N 22/P13/Q10/a,b

10 (a) Explain the importance of feedback in a control system.

[3]

(b) (i) Identify **one** sensor that could be used in a car alarm system.

Justify your choice.

Sensor

Justification

[2]

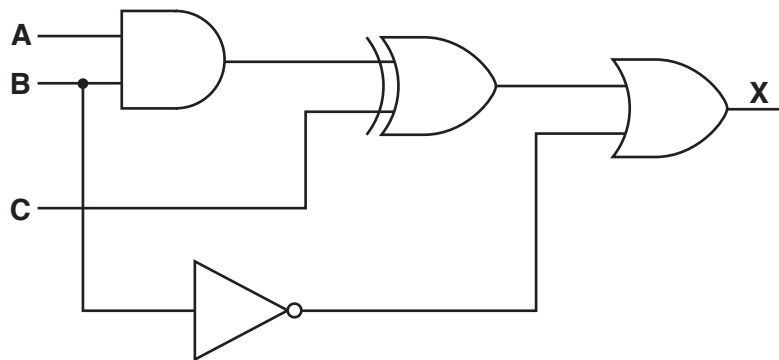
(ii) The car alarm is an example of an embedded system.

Describe the characteristics of an embedded system.

[3]

53. O/N 21/P11/Q3/a,b,c

3 A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

[3]

(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

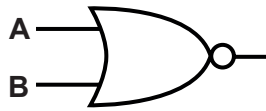
Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

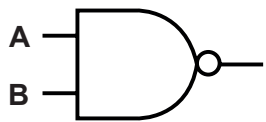
[3]

54. O/N 21/P12/Q2/a,b

- 2 (a) Complete the truth table for each of the following two logic gates.



A	B	Output
0	0	
0	1	
1	0	
1	1	



A	B	Output
0	0	
0	1	
1	0	
1	1	

[2]

- (b) Draw a logic circuit for the following logic expression.

$$X = \text{NOT}(\text{NOT}(A \text{ AND } B) \text{ AND } C)$$



[2]

55. O/N 21/P12/Q3/a

3 Andy likes to play computer games.

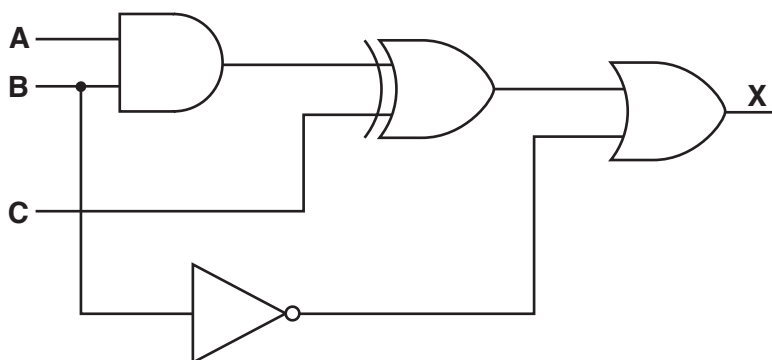
- (a) Andy uses several input devices to play the games. These include a keyboard and a microphone.

Describe the principal operation of a microphone.

[3]

56. O/N 21/P13/Q3/a,b,c

3 A logic circuit is shown:



- (a) Write the logic expression for the logic circuit.

[3]

- (b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

- (c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

[3]

57. M/J 21/P11/Q5/a,b,c

5 Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

[2]

(b) The washing machine's embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine's embedded system.

RAM

ROM

[2]

(c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

[2]

58. M/J 21/P11/Q8

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

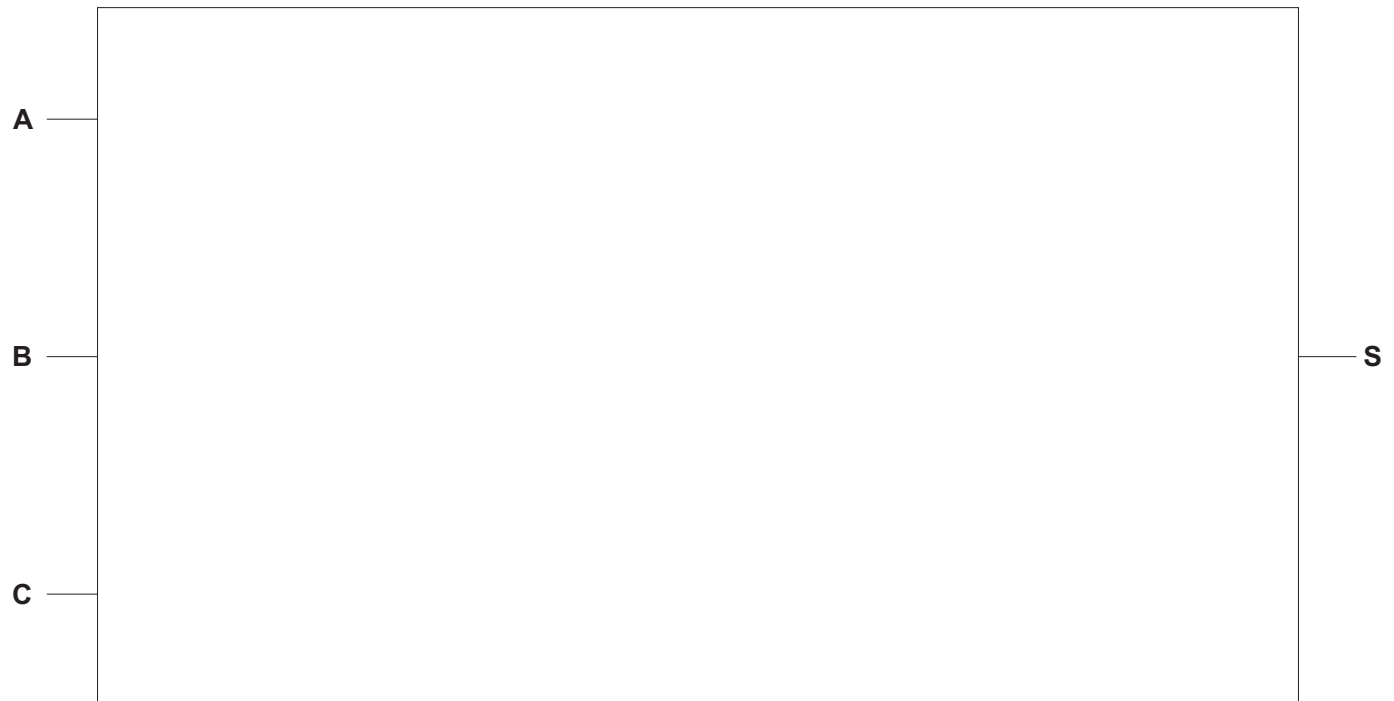
[3]

59. M/J 21/P12/Q3/a,b

3 A logic expression is given:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

(a) Draw the logic circuit for the given expression.



[4]

(b) Complete the truth table for the logic expression:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

A	B	C	Working space	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

60. M/J 21/P13/Q5/a,b,c

5 Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

[2]

(b) The washing machine's embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine's embedded system.

RAM

ROM

[2]

(c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

[2]

61. M/J 21/P13/Q8

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

[3]

Answers Section

1. M/J 26/P11/Q6/a,b

6(a)	1 mark for each correct answer Y = NOT (B OR NOT C) Z = E NOR (NOT C XOR D)	2 Accept any recognized notation Accept Y = B NOR NOT C															
6(b)	1 mark for drawing a truth table with correct inputs 1 mark for the correct outputs in the NAND column <table border="1"><thead><tr><th>Input 1</th><th>Input 2</th><th>NAND</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></tbody></table>	Input 1	Input 2	NAND	0	0	1	0	1	1	1	0	1	1	1	0	2 Ignore extra columns for working
Input 1	Input 2	NAND															
0	0	1															
0	1	1															
1	0	1															
1	1	0															

2. M/J 26/P12/Q6/a,b,c

6(a)	1 mark for each correct point to max 3 - It only has one function, to monitor health - All the hardware such as sensors and a microprocessor is dedicated to the task of monitoring health - The firmware cannot be easily amended or updated because its function is preset	3	
6(b)	No mark for the choice. 1 mark for each correct point to max 2 Monitoring: - The system only records/transmits the data - It does not make use of feedback // it does not change the environment - The output does not affect the next input - It does not contain an actuator	2	Not performing an action is NE for the no actuator mark
6(c)	1 mark for each correct point to max 2 - RAM can be changed; ROM cannot be changed // RAM can be read from and written to, ROM can only be read - RAM is volatile, ROM is non-volatile // RAM is temporary, ROM is permanent - RAM stores currently running data/instructions, ROM stores boot-up/firmware	2	The question asks for differences, so answers must refer to both RAM and ROM Do not accept the expansion of the acronyms, it is given in the question

3. M/J 26/P12/Q8/a,b

8(a)	1 mark for A column and 1 mark for B column <table><thead><tr><th>X</th><th>Y</th><th>Z</th><th>Working space</th><th>A</th><th>B</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td><td>0</td></tr></tbody></table>	X	Y	Z	Working space	A	B	0	0	0		0	0	0	0	1		0	0	0	1	0		0	0	0	1	1		0	1	1	0	0		0	0	1	0	1		0	1	1	1	0		1	0	1	1	1		1	0	2	
X	Y	Z	Working space	A	B																																																				
0	0	0		0	0																																																				
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0	1	0		0	0																																																				
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1	0	0		0	0																																																				
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1	1	0		1	0																																																				
1	1	1		1	0																																																				
8(b)	1 mark for a correct answer NOT (A XOR B) NOT A XOR B // NOT B XOR A NOT (A OR B) OR (A AND B) // (NOT A AND NOT B) OR (A AND B) (NOT A AND NOT B AND NOT C) OR (NOT A AND NOT B AND C) OR (A AND B AND NOT C) OR (A AND B AND C) OR (A AND B AND C) A AND B OR (A NOR B)	1	Accept any correct Boolean notation For example: $\bar{A}.\bar{B}.\bar{C} + \bar{A}.\bar{B}.C + A.B.\bar{C} + A.B.C$ Or $A'B'C' + A'B'C + ABC' + ABC$																																																						

4. M/J 26/P13/Q2/a,b

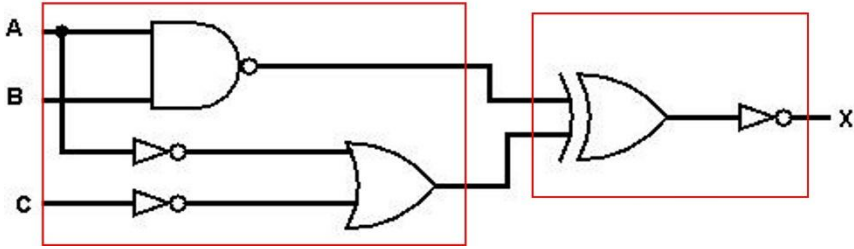
2(a)	1 mark for each correct column					2		
	X	Y	Z	Working space	A			B
	0	0	0		1			1
	0	0	1		0			0
	0	1	0		1			0
	0	1	1		1			1
	1	0	0		1			1
	1	0	1		1			0
	1	1	0		1			0
	1	1	1		1			1
2(b)	1 mark for two correct					1	Both are needed.	
	NAND and OR							

5. M/J 26/P13/Q8/a,b,c

8(a)	1 mark for name, 1 mark for a matching description - EPROM - Erase using UV rays // needs removing to erase // have to erase whole device // supports many read/write cycles - EEPROM - Erase using electric signals // erase in situ // can update parts of memory // uses floating gate technology // supports many read/write cycles	4	No mark for expanding acronym for description.
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8(b)	1 mark each. - The microphone has a diaphragm / ribbon - The incoming sound waves cause vibrations of the diaphragm ... causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone). // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) - An electrical signal is produced	3	For MP1 accept diaphragm equivalent e.g. cone.
8(c)	1 mark each - Temporary storage of data ...until the device is ready to receive it // until a sound player takes the data - Data may be sent faster than it can be received // Allows for different speed of transmission between devices - Allows processor to carry out other tasks	3	

6. O/N 25/P11/Q3/a,b

3(a)	One mark per bullet point, max 2 marks (A NAND B) and (NOT A OR NOT C) - XOR and NOT 	2
3(b)	1 mark per bullet point, max 2 marks - One correct term: (R AND S AND NOT T) // (NOT R AND NOT S AND T) - Second correct term plus the OR in correct place Answer: Q = (R AND S AND NOT T) OR (NOT R AND NOT S AND T)	2

7. O/N 25/P11/Q6/c

6(c)	1 mark per bullet point, max 3 marks - DRAM requires to be refreshed/charged SRAM does not request a refresh - DRAM stores each bit as charge SRAM uses flip-flop to store each bit - DRAM is less expensive to manufacture SRAM is more expensive to manufacture - DRAM has slower access speeds SRAM has faster access times - DRAM has higher <u>storage/bit/data density</u> SRAM has lower storage/bit/data density - DRAM is used in main memory SRAM is used in cache	3
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8. O/N 25/P11/Q9

9	1 mark per bullet point, max 3 marks - Monitoring systems do not take any action whereas control systems act autonomously to change the environment if the values are out of a prescribed range - Control systems use actuators monitoring systems do not have any actuators - Control systems make use of feedback monitoring systems do not make use of feedback - The output from a monitoring system does not affect the subsequent input whereas output from a control system affects the next input.	3
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9. O/N 25/P11/Q10/a,b

10(a)	1 mark per bullet point, max 3 marks Generic mark points: (max 3) - Additive manufacturing - Uses a digital file created from 3D modelling software or Computer Aided Design (CAD) software - The printer builds up the model one layer at a time - The process is repeated for each layer - Some materials may need time to cool and set or UV curing using LEDs for photopolymer resins Specific mark points: (max 1) e.g. - Fused Deposition Modelling (FDM) // Material is heated and pushed through nozzle / extruder - Stereolithography (SLA) // Photosensitive liquid resin is exposed to a UV-laser beam - Digital Light Processing (DLP) // Photosensitive liquid resin is exposed to a UV projected image of the layer to cure the resin - Selective Laser Sintering (SLS) // Uses a laser to form objects from powdered material, the material can be plastics or metals	3
10(b)	1 mark - To free up the processor to carry out other tasks as the rate data is received by the printer is different from the rate at which it can be processed - To manage the mismatch in speed between the processor and the peripheral / 3D printer	1

10. O/N 25/P12/Q8/b(i)

8(b)(i)	1 mark per bullet point, max 4 marks Methods to determine point of touch (Max 2) • Resistive // two layers of material make contact and complete a circuit • Capacitive // contact with the screen creates a change in charge • Infrared // the beams are broken by the pointing device • Optical imaging // a shadow is created when the surface is touched • Acoustic pulse // the acoustic wave is absorbed by the pointing device • Selection from the menu (Max 2) • The point of touch is used to determine the x and y coordinates • The menu item corresponding to the coordinate position is recognised and the menu item is added to the order	4
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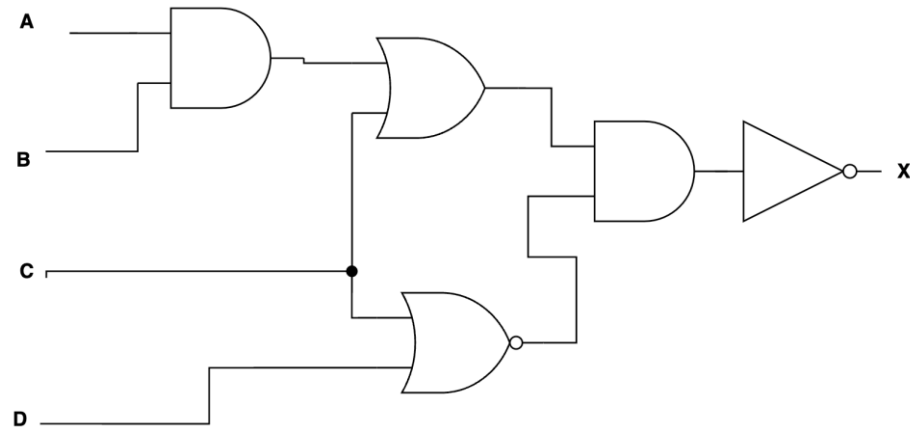
11. O/N 25/P12/Q9/a,b

9(a)	1 mark per bullet point, max 2 marks • NOT (A XOR B) • NAND ((B AND C) OR A) X = (NOT (A XOR B)) NAND ((B AND C) OR A) // X = ((B AND C) OR A) NAND (NOT (A XOR B))	2
9(b)	1 mark for 1 or 2 row numbers correct, 2 marks all three correct Row 2 Row 4 Row 7	2

12. O/N 25/P12/Q11

11	1 mark per bullet point, max 3 marks Identification: Control system (no mark) Justification: • The system uses an actuator to open or close the doors • The output / opening the doors allows people to move, which changes the input to the sensor • The system acts autonomously on the feedback from a sensor	3
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13. O/N 25/P13/Q3/a,b

3(a)	1 mark per shaded area, max 2 marks <table><tr><td>A</td><td>B</td><td>C</td><td></td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C		X	0	0	0		1	0	0	1		0	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C		X																																											
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0	0	1		0																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											
3(b)	1 mark per bullet point, max 2 marks - A AND B OR C - C NOR D and final two gates 	2																																													

14. O/N 25/P13/Q4/c

4(c)	One mark for each correct row	3								
<table><tr><th>Description</th><th>Memory Technology</th></tr><tr><td>A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.</td><td>EEPROM</td></tr><tr><td>A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.</td><td>EPROM</td></tr><tr><td>A read only memory where the contents can be written only once after manufacture.</td><td>PROM</td></tr></table>			Description	Memory Technology	A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	EEPROM	A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	EPROM	A read only memory where the contents can be written only once after manufacture.	PROM
Description	Memory Technology									
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	EEPROM									
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	EPROM									
A read only memory where the contents can be written only once after manufacture.	PROM									

15. O/N 25/P13/Q7/d

7(d)	1 mark per reason, 1 mark for corresponding justification x2 max 4 marks • Lower cost per unit of storage • ... so the high capacity of storage required for large number of video files will be less costly • A large number of read/write operations are being performed continuously • ... and magnetic storage is likely to have a longer life span compared with solid state	4
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16. M/J 25/P11/Q1/a,b

1(a)	1 mark each: $X = Q \text{ AND } (\text{NOT } P \text{ OR } R)$ $Y = S \text{ XOR } (Q \text{ NOR } R)$	2
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1(b)	1 mark for first 4 rows, 1 mark for second 4 rows					2
A	B	C	Working space	X		
0	0	0		1		
0	0	1		1		
0	1	0		0		
0	1	1		1		
1	0	0		0		
1	0	1		0		
1	1	0		1		
1	1	1		1		

17. M/J 25/P11/Q4/b

4(b)	1 mark for sensor, 1 mark for corresponding use e.g. <table><tr><th>Sensor</th><th>Use</th></tr><tr><td>Pressure</td><td>Detects when the pressure of an item is removed from a shelf / put back on a shelf</td></tr><tr><td>Infrared</td><td>Detects when the beam is broken for an item removed / added</td></tr></table>	Sensor	Use	Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf	Infrared	Detects when the beam is broken for an item removed / added	2
Sensor	Use							
Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf							
Infrared	Detects when the beam is broken for an item removed / added							

18. M/J 25/P11/Q6/a(i)

6(a)(i)	1 mark for each correctly completed space • platters • spindle • read/write head • magnetic field The magnetic hard disk has one or more platters that can be magnetised. These are mounted on a spindle and rotate at high speed. A read/write head is moved across the surface on an arm. When data is read the changes in the magnetic field produce a change in the electric current.	4
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19. M/J 25/P12/Q4/a,b

4(a)	1 mark each: • $X = (R \text{ NOR } S) \text{ XOR } T$ • $Y = (T \text{ AND } U) \text{ OR } U$	2																																													
4(b)	1 mark for first four rows, 1 mark for second four rows <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		0	1	1	0		0	1	1	1		1	2
A	B	C	Working space	X																																											
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0	0	1		1																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		0																																											
1	1	0		0																																											
1	1	1		1																																											

20. M/J 25/P13/Q4/a,b

4(a)	1 mark for each correct statement $X = B \text{ NOR } (A \text{ NOR } C)$ $Y = (B \text{ AND } C) \text{ XOR NOT } D$	2																																													
4(b)	1 mark for 1st 4 rows 1 mark for 2nd 4 rows <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C	Working space	X																																											
0	0	0		1																																											
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0	1	1		0																																											
1	0	0		0																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

21. O/N 24/P11/Q3/c,d

3(c)	1 mark for each bullet point (max 2) e.g. • DRAM requires constant refresh cycles unlike SRAM • DRAM has lower access speed than SRAM	2
3(d)(i)	1 mark for each bullet point (max 4) • The disc is spun at high speed • A laser is shone onto the disc to read / write • ... using optical head to move it into position • ... it follows the spiral track from the centre outwards • When writing the laser burns pits to represent the data • When reading the laser reflects from pits and lands • The reflection from a pit and a land is different • ... the differences are interpreted as 1 or 0	4
3(d)(ii)	1 mark for each bullet point (max 3) • The computer and optical disc reader/writer send and receive at different speeds • A buffer allows temporary storage of the data • ... so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer // so the computer can carry on with other tasks • ... so the optical disc reader/writer is not overloaded • ... and so that data is transferred to optical disc reader/writer from the buffer at the slower rate	3

22. O/N 24/P11/Q5/a,b,c

5(a)	1 mark for all 3 appropriate sensors	1								
	<table><tr><th>Scenario</th><th>Type of sensor</th></tr><tr><td>A door is open.</td><td>Pressure sensor // Infra-red</td></tr><tr><td>The external daylight-level is below a set amount.</td><td>Light sensor</td></tr><tr><td>A person is detected within 2 metres.</td><td>Infra-red</td></tr></table>	Scenario	Type of sensor	A door is open.	Pressure sensor // Infra-red	The external daylight-level is below a set amount.	Light sensor	A person is detected within 2 metres.	Infra-red	
Scenario	Type of sensor									
A door is open.	Pressure sensor // Infra-red									
The external daylight-level is below a set amount.	Light sensor									
A person is detected within 2 metres.	Infra-red									
5(b)	1 mark for each correct expression X = E AND A AND C Y = E AND (B OR C OR D)	2								

5(c)	1 mark for each bullet point (max 3) for a correct justification No marks for the identification of the system Monitoring system - There is no use of feedback // The light and the alarm are just warnings - The output of the floodlight or the audio alarm does not affect the input of data from the sensors - The system does not have any actuators	3
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23. O/N 24/P12/Q1/a,b

1(a)	1 mark for each shaded part. <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	1	0	1	0	1	0	1	1	1	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1	1	2
A	B	C	X																																			
0	0	0	1																																			
0	0	1	1																																			
0	1	0	1																																			
0	1	1	1																																			
1	0	0	1																																			
1	0	1	0																																			
1	1	0	1																																			
1	1	1	1																																			
1(b)	1 mark for (P XOR Q) and (Q OR NOT R) 1 mark for second XOR gate and final NAND gate with appropriate inputs	2																																				

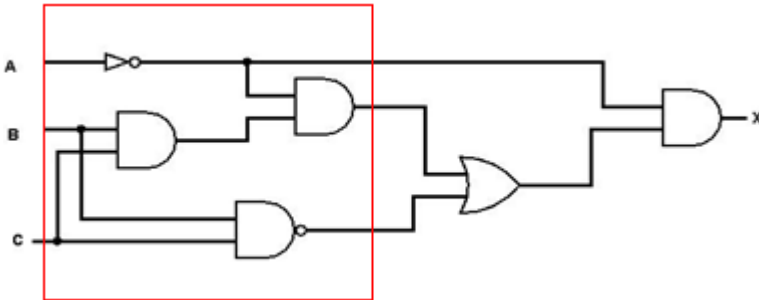
24. O/N 24/P12/Q2/a,b,c

2(a)	1 mark for each bullet point (max 3) e.g. • It is difficult to change / update the firmware by the user • Errors cannot be fixed easily // Troubleshooting / fault-finding / repairing is a specialist task / expensive • Functionality cannot be changed/extended easily // Cannot be easily adapted for another task • Faulty / outdated devices are often thrown away rather than repaired • ... leading to e-waste	3
2(b)	1 mark from: • DRAM is less expensive to manufacture/purchase than SRAM in the embedded system • DRAM has a higher bit density per chip // more data can be stored per chip	1
2(c)	1 mark for each bullet point (max 2) • EPROM uses ultraviolet light to erase data whilst EEPROM uses an electrical signal to do this • EPROM has to be removed from the circuit board when changing the data whilst EEPROM remains in the circuit when the data is changed • EPROM erases all the data, EEPROM can erase parts of the data	2

25. O/N 24/P12/Q9/a,b

9(a)	1 mark for sensor and its use in this system • Infra-red sensor • Measure / check the height of the vehicle • Pressure sensor • Measure / check the weight of the vehicle	2
9(b)	1 mark for each bullet point (max 2) for a correct justification No marks for the identification of the system Monitoring system • Because there is no use of feedback // the warning sign is only an indicator • ... the output of the turning on of the sign does not affect the input of data from the sensors • The system does not have any actuators	2

26. O/N 24/P13/Q1/a,b,c

1(a)	1 mark for: Output is only 1 if one input is 1 and the other is 0 // Output is only 1 if both inputs are different // Output is only 0 if both inputs are the same	1
1(b)	1 mark each: • $((A \text{ NOR } B) \text{ AND } (A \text{ OR } B))$ • $\text{NAND } (B \text{ AND } C)$ $X = ((A \text{ NOR } B) \text{ AND } (A \text{ OR } B)) \text{ NAND } (B \text{ AND } C)$	2
1(c)	1 mark for NOT gate, 2 correct AND gates and the NAND gate 1 mark for OR gate and the final AND gate 	2

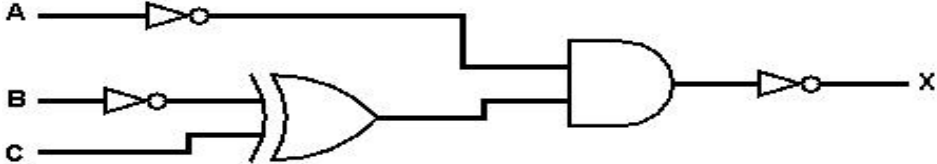
27. O/N 24/P13/Q2/a,b

2(a)	1 mark for each difference e.g. • DRAM requires constant refreshing, but SRAM does not need to be constantly refreshed • DRAM generally has greater capacities per chip, but SRAM generally has lower capacities per chip • DRAM has slower access times than SRAM	2
2(b)	1 mark for each bullet point (max 4) • The computer and the hard disk drive transmit and receive at different speeds // The computer transfers data faster than the HDD can receive • The buffer is used for temporary storage • ... so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer • ... and so that data is transferred to hard disk drive from the buffer at the slower rate	4

28. O/N 24/P13/Q3

3	1 mark for each bullet point (max 3) - The system uses feedback - The system acts independently - Input data causes the car to brake ... braking decreases the/changes the distance to the object ... this new distance is used to determine whether/how hard the car needs to brake	3
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29. M/J 24/P11/Q1/a,b

1(a)	1 mark for: $(A \text{ XOR } B) \text{ NOR } C$	1
1(b)	1 mark for NOT B XOR C 1 mark for NOT A and final AND plus NOT 	2

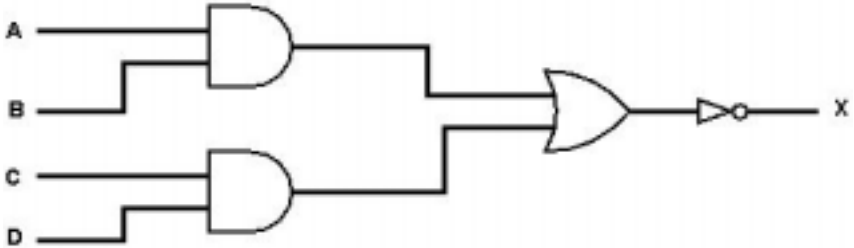
30. M/J 24/P11/Q2/a,b,c

2(a)	1 mark each to max 2 : - The doorbell only performs the specific tasks of motion detection/video recording/doorbell ringing - The motion sensor and digital camera are built into the doorbell - The CPU/memory/storage/software are all dedicated to this task only - Only a dedicated microprocessor is required due to the limited processing requirements	2
2(b)	No mark for identification of monitoring or control 1 mark each to max 2 for justification: Monitoring: - The turning on of the digital camera does not affect the input to the sensor/button - The transmission of the data/video does not affect the input to the sensor/button - The ringing of the doorbell does not affect the input to the button Control: - Video doorbell does not only store the values from the motion sensor - The data is processed, generating a signal to start the digital camera recording - Button pressed/motion detected causes a signal to be sent over a network to the smartphone	2

2(c)(i)	1 mark each to max 2: • Current reading/data from motion sensor • Current/recent video • Instructions being executed • Start-up/BIOS/boot-up instructions	2										
2(c)(ii)	1 mark for each row: <table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The two types of logic gate that can be used to create solid state devices</td><td>NAND NOR</td></tr><tr><td>The number of transistors contained in each cell</td><td>2</td></tr><tr><td>The type of gate that can retain electrons without power</td><td>floating</td></tr><tr><td>The type of gate that allows or stops current from passing through</td><td>control</td></tr></table>	Statement	Answer	The two types of logic gate that can be used to create solid state devices	NAND NOR	The number of transistors contained in each cell	2	The type of gate that can retain electrons without power	floating	The type of gate that allows or stops current from passing through	control	4
Statement	Answer											
The two types of logic gate that can be used to create solid state devices	NAND NOR											
The number of transistors contained in each cell	2											
The type of gate that can retain electrons without power	floating											
The type of gate that allows or stops current from passing through	control											
2(c)(iii)	1 mark each to max 2: • Captured video is transmitted to buffer • ... video is transmitted from buffer to smartphone • Store recent data in a buffer for the user to rewind • ... instead of storing everything in secondary storage • Store readings from motion sensor • ... until the microprocessor can process them • Store video from digital camera • ... before moving it to secondary storage	2										

31. M/J 24/P12/Q1/a,b

1(a)	1 mark for each correct answer: <i>NAND</i> - The output is 0 when both inputs are 1, otherwise the output is 1 <i>NOR</i> - The output is 1 when both inputs are 0, otherwise the output is 0 <i>XOR</i> - The output is 1 when one of the inputs is 1 and the other input is 0, otherwise the output is 0 <i>OR</i> - The output is 0 when both inputs are 0, otherwise the output is 1	4
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1(b)	1 mark for both AND gates with correct inputs 1 mark for correct OR and NOT gates with correct inputs and no superfluous gates: 	2
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32. M/J 24/P12/Q2/a,b,c

2(a)	1 mark for each correctly completed statement: • (LCD) displays/screens/lenses • gyroscope/accelerometer • direction/speed • digital cameras A headset can have one or two (LCD) displays/screens/lenses that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor. This sensor is a gyroscope/accelerometer. The data is transmitted to a microprocessor that analyses the data to identify the direction/speed of movement. Some headsets use digital cameras that record the user's eye movements for analysis.	4
2(b)	1 mark each to max 3: • The buffer is used as a temporary store for data going to the headset • Data is transferred into the buffer by the computer • Data is retrieved from the buffer by the headset • When the buffer is empty/full an interrupt is sent to the computer requesting more data/stopping further data being sent • When the headset has enough data/needs more data, an interrupt is sent by the headset to the computer to stop sending data from buffer	3
2(c)	1 mark each to max 3: • EEPROM allows frequent/multiple read/write/erase operations • ... so the headset can take advantage of new features • ... without fully erasing the contents of the firmware in the headset first // can erase a particular byte or the whole EEPROM • ... without removing the chip(s)/firmware from the headset • ... the contents of the firmware in the headset can be changed by the user without technical expertise • Cheaper to manufacture so headset will be cheaper to purchase	3

33. M/J 24/P13/Q6

6

1 mark for each correct line:

3

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Logic expression

NOT (A XOR B) AND C

(A OR C) AND NOT B

A NAND B NAND C

(A NAND B) OR C

NOT (A AND B AND C)

34. M/J 24/P13/Q7/a,c,d

7(a)	1 mark for sensor and matching purpose to max 2: <table><tr><th>Sensor</th><th>Purpose of sensor in navigation system</th></tr><tr><td>Pressure</td><td>To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on</td></tr><tr><td>Infra-red</td><td>To detect if there is an obstacle in the way // to indicate that it has reached the desired table</td></tr><tr><td>Sound</td><td>To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop</td></tr></table>	Sensor	Purpose of sensor in navigation system	Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on	Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table	Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop	2
Sensor	Purpose of sensor in navigation system									
Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on									
Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table									
Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop									
7(c)	1 mark each to max 2: • Feedback ensures that a system operates within set criteria / constraints • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be automatically adjusted	2								

7(d)	1 mark each to max 4 : - Resistive: The space between the conductive layers is removed/the layers touch and a circuit is completed - Capacitive: The electrical charge changes where the user pressed - The point of contact is identified ... from the change in electrical field - The software/microprocessor calculates the coordinates	4
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35. O/N 23/P11/Q4/a,b

4(a)	1 mark for each set of 4 rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1	1	2
A	B	C	X																																			
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1	1	0	1																																			
1	1	1	1																																			
4(b)	1 mark for both AND gates 1 mark for NOT gate <u>and</u> OR gate <u>and</u> NOR gate <pre>graph LR; A --- AND1[AND]; B --- OR1[OR]; C --- OR1; C --- NOR1[NOR]; OR1 --- NOT1[NOT]; AND1 --- AND2[AND]; NOT1 --- AND2; NOR1 --- AND2; AND2 --- X[X]</pre>	2																																				

36. O/N 23/P11/Q7/a,b,c

7(a)	1 mark for each bullet point (max 3) Generic mark points: • Additive manufacturing • Uses a digital 3D model or a Computer Aided Design (CAD) (file) • Builds up the model one layer at a time • ...starting from the bottom • ...using x, y and z co-ordinates • The material is fused / cured together layer by layer Specific mark points: Fused Deposition Modelling (FDM) • Material is heated and pushed through nozzle / extruder Stereolithography (SLA) • Photosensitive liquid resin is exposed to a UV-laser beam Digital Light Processing (DLP) • Uses liquid plastic resin melted with arc lamps Selective Laser Sintering (SLS) • Uses a laser to form objects from powdered material	3
7(b)	1 mark for each bullet point (max 2) • To prevent overheating // ensure material is hot enough • ...by identifying the temperature of the object (being printed) • ...by identifying the temperature of the material being used	2
7(c)	1 mark for each bullet point (max 2) • Dynamic RAM has lower cost per unit • A fast access speed is not needed • Higher bit density // more data can be stored per chip	2

37. O/N 23/P11/Q9/a,b

9(a)	One mark for each bullet point (max 2) • Feedback ensures that a system operates within set criteria / constraints • ...by enabling system output to affect (subsequent) system input • ...thus allowing conditions to be <u>automatically</u> adjusted	2
9(b)	One mark for each reason to max 3 to match the example given • Dedicated to one task applied to example • Does not require much processing power applied to example • Built into a larger system applied to example • Contains firmware that cannot be easily updated applied to example • The system does not have its own operating system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	3

38. O/N 23/P12/Q1/a,b,c

1(a)	1 mark for: infra-red / proximity (sensor)	1
1(b)	1 mark for each bullet point (max 2) - Actuator generates a signal / causes an action / converts electrical energy into a mechanical force ... to push an arm // to open a trap door // to pick up the chocolate bar with the incorrect weight	2
1(c)(i)	1 mark for each feature (max 2) - Dedicated to a single task // limited number of functions - Built into a larger system // integrated into a larger system - An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	2
1(c)(ii)	1 mark for each bullet point (max 1) - Difficult to change / update the firmware by the user // Difficult to upgrade devices to take advantage of new technology - Cannot be easily adapted for another task - Troubleshooting faults/upgrading is a specialist task - Devices are often thrown away rather than repaired as difficult to upgrade or repair	1

39. O/N 23/P12/Q4/a,b

4(a)	1 mark for each bullet point - A NAND B - NOT(B XOR C) - Final NAND $X = A \text{ NAND } B \text{ NAND } (\text{NOT}(B \text{ XOR } C))$	3
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4(b)	1 mark for each set of rows (shaded)	2																																				
<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>			A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	1	1	1	0	1	1	1	1	1
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40. O/N 23/P13/Q4/a,b

4(a)	1 mark for each set of highlighted rows	2																																				
<table><tr><th>P</th><th>Q</th><th>R</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>			P	Q	R	Y	0	0	0	1	0	0	1	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0
P	Q	R	Y																																			
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1	1	1	0																																			

4(b)	1 mark for P AND Q and NOT Q OR R 1 mark for NOT gate and AND gate and XOR gate	2

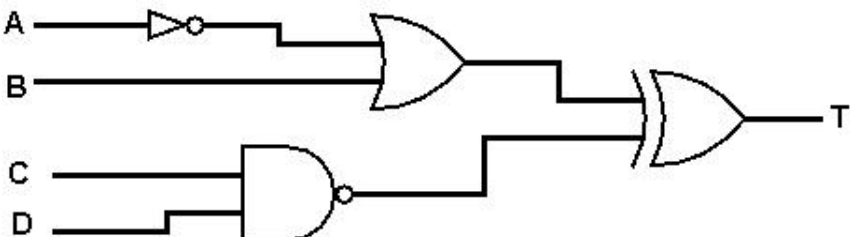
41. O/N 23/P13/Q7/a,b,c

7(a)	1 mark for each bullet point (max 2). - Static RAM has faster access time ...because it does not need to be refreshed ...used on the CPU for improvement of CPU cache speed - Static RAM has lower	2
7(b)	1 mark for each bullet point (max 2) - EEPROM allows frequent / multiple read / write / erase operations ...which means that the contents of the firmware in the headset can be changed easily ...without fully erasing the contents of the firmware in the headset first	2
7(c)	1 mark for each bullet point (max 4) - The print instructions and data are sent by the laptop to a buffer (at laptop speed) - The data is transferred from the buffer to the printer (at printer speed) ...allowing user to continue using the laptop // and allowing processor to continue processing ...instead of waiting for relatively slower printer - When the buffer is empty an interrupt is sent to the laptop ...requesting more data	4

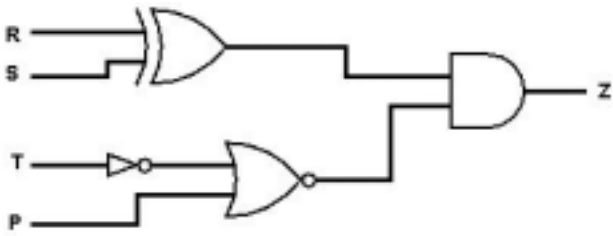
42. M/J 23/P11/Q4/a,b

4(a)(i)	1 mark each to max 2 - Stores the bootstrap program // start-up instructions for the central computer // BIOS - Stores the start-up instructions for the CCTV system/cameras // firmware for CCTV - Stores the kernel of the Operating System // stores parts of the Operating System	2
4(a)(ii)	1 mark each to max 2 - Costs less per unit - Higher storage density - Simple design – uses fewer transistors	2
4(b)	1 mark for reason, 1mark for application/justification - The computer will have a large number of read/write operations because it is working all the time .. magnetic storage has more longevity - Magnetic storage costs less per storage unit ... videos are large files and therefore very large storage capacity is required	4

43. M/J 23/P11/Q5/a,b

5(a)	1 mark for 2 gates 2 marks for all 4 gates 	2
5(b)	1 mark each NAND 0 is only output when both inputs are 1 // 1 is only output when none, or (either) one of the inputs is 1 NOR 1 is only output when both inputs are 0 // 0 is only output when (either) one or both inputs are 1	2

44. M/J 23/P12/Q6/a,b

6(a)	1 mark for correct XOR and AND gates, with correct inputs 1 mark for correct NOT and NOR gates with correct inputs 	2
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6(b)

1 mark for first 4 rows
1 mark for last 4 rows

P	Q	R	Working space	Z
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		0
1	1	0		1
1	1	1		1

2

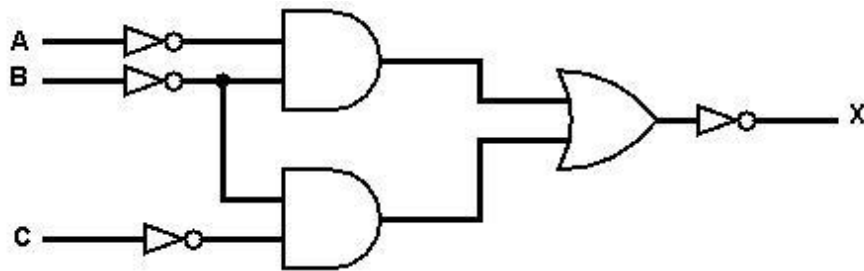
45. M/J 23/P13/Q1/a,b

1(a)	NOT B	1																																				
1(b)	1 mark for first 4 rows correct; 1 mark for second 4 rows correct	2																																				
<table><tr><td>A</td><td>B</td><td>C</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>			A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	0	1	1	1	1
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46. M/J 23/P13/Q3/a

3(a)	1 mark for each term • Resistive • Circuit • Capacitive • Change • Coordinates A resistive touchscreen has two layers. When the user touches the screen, the layers touch and a circuit is completed. A capacitive touchscreen has several layers. When the top layer is touched there is a change/increase/decrease in the electric current. A microprocessor identifies the coordinates of the touch.	5
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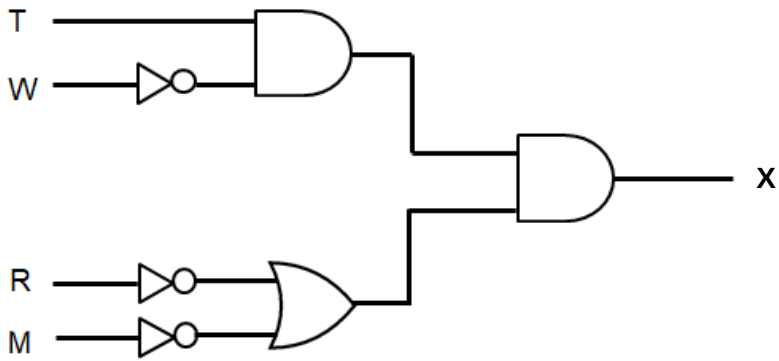
47. O/N 22/P11/Q3/a,b

3(a)	1 mark for each bullet point: • NOT A AND NOT B and NOT B AND NOT C // A NOR B and B NOR C • final OR and NOT gates (with correct inputs) // NOR gate (with correct inputs) 	2																																				
3(b)	1 mark for each set of rows as highlighted: <table><thead><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></tbody></table>	A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	1	1	0	0	0	1	0	1	1	1	1	0	1	1	1	1	1	2
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48. O/N 22/P11/Q9/a,b

9(a)	1 mark for each bullet point (max 2): - the embedded system is built into / integrated into the TV - combination of hardware and software designed for a specific function - The system is not easily changed/updated by the TV owner	2
9(b)	1 mark for each benefit and 1 mark for corresponding expansion (max 2). For example: - no additional equipment is needed to change ... enables firmware updates by non-technical users - can be erased and reprogrammed several times ... so firmware can be updated ... can erase a particular byte or the whole EEPROM - possible to reprogram / update ... without removing it from the device	2

49. O/N 22/P12/Q3/a,b,c

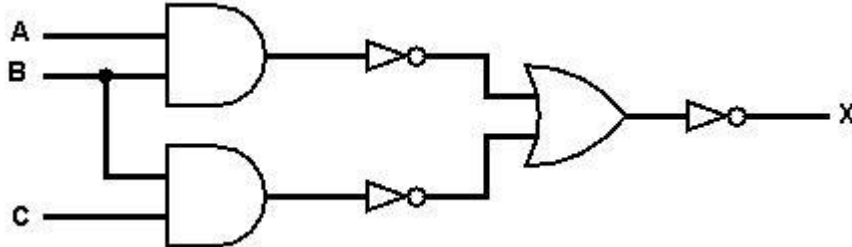
3(a)	1 mark for T AND NOT W 1 mark for NOT R OR NOT M 1 mark for final AND 	3
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3(b)	1 mark for each set of rows as highlighted:	2																																				
<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>			A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0
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3(c)	1 mark for each bullet point: - to store data that does not change - data must be stored even when device is without power - to store boot up instructions / system software / firmware / BIOS	2																																				

50. O/N 22/P13/Q1/c

1(c)	1 mark for purpose (max 2): Purpose: - to act as temporary storage // to store (downloaded) data ... before it is used by the receiving device ... to allow processes / devices to operate at different speeds // independently of each other 1 mark for each example (max 1): Examples: - printer buffer used when data is transferred from a computer to a printer - video buffer when streaming videos - keyboard buffer when performing data entry	3
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51. O/N 22/P13/Q5/a,b

5(a)	1 mark for each bullet point: • NOT (A AND B) • NOT (B AND C) • NOT(NOT(A AND B) OR NOT(B AND C)) 	3																																				
5(b)	1 mark for each set of highlighted rows. <table data-bbox="280 761 839 1348"><thead><tr><th>P</th><th>Q</th><th>R</th><th>Y</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></tbody></table>	P	Q	R	Y	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	1	1	0	0	0	1	0	1	0	1	1	0	1	1	1	1	0	2
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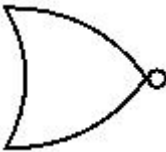
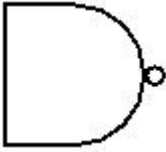
52. O/N 22/P13/Q10/a,b

10(a)	1 mark for each bullet point: • to ensure the system operates with the given criteria • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be <u>automatically</u> adjusted	3
10(b)(i)	1 mark for identification of a suitable sensor 1 mark for corresponding justification Example: • sound sensor • if a sound occurs inside the car the alarm is activated • infra-red sensor • senses the heat of person in the car / infra-red beams are broken • pressure sensor • an intruder sits in the driver's seat	2

10(b)(ii)	1 mark for each bullet point (max 3): - the embedded system is built into / integrated (into the car alarm) - combination of hardware and software designed for a specific function - must have a processor, memory and input / output - The system is not easily changed/updated by the car owner	3
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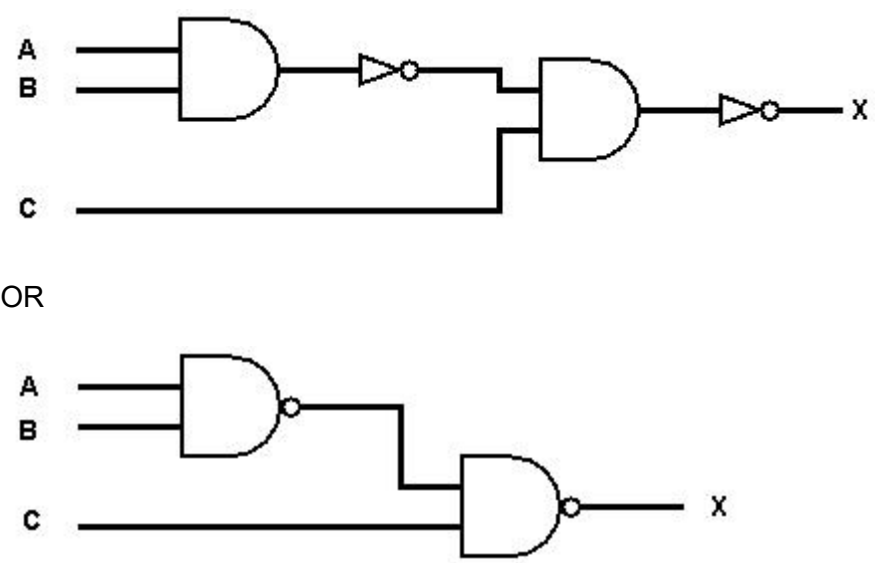
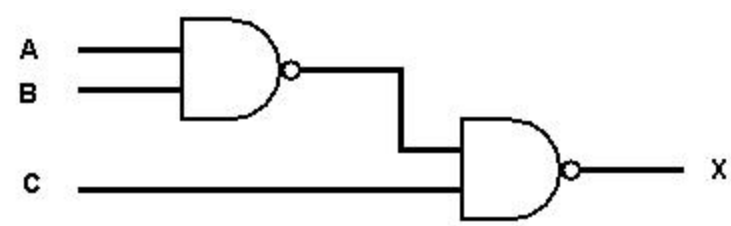
53. O/N 21/P11/Q3/a,b,c

3(a)	1 mark per bullet point • A AND B ... • ... XOR C ... • ... OR NOT B ((A AND B) XOR C) OR NOT B	3																																													
3(b)	1 mark for each set of 4 rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
A	B	C	Working space	X																																											
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3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR  <table border="1" data-bbox="683 309 1150 633"> <thead> <tr> <th>A</th><th>B</th><th>OUTPUT</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND  <table border="1" data-bbox="678 701 1145 1025"> <thead> <tr> <th>A</th><th>B</th><th>OUTPUT</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	OUTPUT	0	0	1	0	1	0	1	0	0	1	1	0	A	B	OUTPUT	0	0	1	0	1	1	1	0	1	1	1	0	3
A	B	OUTPUT																														
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54. O/N 21/P12/Q2/a,b

2(a)	1 mark for each completely correct truth table NOR <table border="1" data-bbox="288 1243 619 1570"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND <table border="1" data-bbox="288 1637 619 1964"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0	A	B	Output	0	0	1	0	1	1	1	0	1	1	1	0	2
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A	B	Output																														
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0	1	1																														
1	0	1																														
1	1	0																														

2(b)	1 mark for each correct bullet point • NOT (A AND B) // A NAND B • NOT the result AND C // the result NAND C  OR 	2
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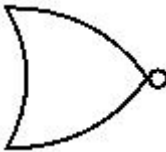
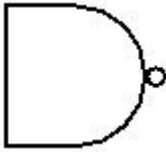
55. O/N 21/P12/Q3/a

3(a)	1 mark for each bullet point to max 3 • The microphone has a diaphragm / ribbon • The incoming sound waves cause vibrations of the diaphragm • ... causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone) // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) • An electrical signal is produced	3
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56. O/N 21/P13/Q3/a,b,c

3(a)	1 mark per bullet point • A AND B ... • ... XOR C ... • ... OR NOT B ((A AND B) XOR C) OR NOT B	3
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3(b)	1 mark for each set of 4 rows (shaded) <table><tr><td>A</td><td>B</td><td>C</td><td>Working space</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
A	B	C	Working space	X																																											
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1	1	1		0																																											

3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> NAND  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	OUTPUT	0	0	1	0	1	0	1	0	0	1	1	0	A	B	OUTPUT	0	0	1	0	1	1	1	0	1	1	1	0	3
A	B	OUTPUT																														
0	0	1																														
0	1	0																														
1	0	0																														
1	1	0																														
A	B	OUTPUT																														
0	0	1																														
0	1	1																														
1	0	1																														
1	1	0																														

57. M/J 21/P11/Q5/a,b,c

5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
5(b)	1 mark for RAM, 1 mark for ROM RAM: Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: Store the start-up instructions (for the washing cycles)	2
5(c)	1 mark per bullet point - The system uses feedback - The system causes the temperature to change // produces an action	2

58. M/J 21/P11/Q8

8	1 mark per correct row	3																								
<table><tr><th>Statement</th><th>AND</th><th>NAND</th><th>NOR</th><th>XOR</th><th>OR</th></tr><tr><td>The output is 1 only when both inputs are 1</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>The output is 1 only when both inputs are different</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>The output is 1 only when both inputs are 0</td><td></td><td></td><td>✓</td><td></td><td></td></tr></table>			Statement	AND	NAND	NOR	XOR	OR	The output is 1 only when both inputs are 1	✓					The output is 1 only when both inputs are different				✓		The output is 1 only when both inputs are 0			✓		
Statement	AND	NAND	NOR	XOR	OR																					
The output is 1 only when both inputs are 1	✓																									
The output is 1 only when both inputs are different				✓																						
The output is 1 only when both inputs are 0			✓																							

59. M/J 21/P12/Q3/a,b

3(a)	1 mark for each correct gate, with correct inputs <pre>graph LR; A --- AND1[AND]; B --- AND1; B --- AND2[AND]; C --- AND2; C --- OR1[OR]; AND1 --- AND2; AND2 --- OR2[OR]; OR1 --- OR2; OR2 --- S</pre>	4																																													
3(b)	1 mark for each half (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>S</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	S	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C	Working space	S																																											
0	0	0		0																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		0																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

60. M/J 21/P13/Q5/a,b,c

5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
5(b)	1 mark for RAM, 1 mark for ROM RAM: Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: Store the start-up instructions (for the washing cycles)	2

5(c)	1 mark per bullet point - The system uses feedback - The system causes the temperature to change // produces an action	2
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61. M/J 21/P13/Q8

8	1 mark per correct row	3																								
<table><tr><th>Statement</th><th>AND</th><th>NAND</th><th>NOR</th><th>XOR</th><th>OR</th></tr><tr><td>The output is 1 only when both inputs are 1</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>The output is 1 only when both inputs are different</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>The output is 1 only when both inputs are 0</td><td></td><td></td><td>✓</td><td></td><td></td></tr></table>			Statement	AND	NAND	NOR	XOR	OR	The output is 1 only when both inputs are 1	✓					The output is 1 only when both inputs are different				✓		The output is 1 only when both inputs are 0			✓		
Statement	AND	NAND	NOR	XOR	OR																					
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